Causal Graphs and Optimization

IA pour la Science

February 16$^{\text{th}}$, 2026

Questions to discuss

1. There is a difference between the causal graphs G_t and G_{t-1} for insufficient T.
2. (Th.) There is no difference between the causal graphs G_t and G_{t-1} for sufficient T.
3. The exhaustive and combinatorial algorithms are special cases of optimization.
4. The latent space is a spectrum of the observable domain $t \to s$ as in Laplace transform.
5. The time is continuous as in the Laplace transform with adaptation to the neural ODE.

MVP

1. Discrete time.
2. Latent space.
3. Direct and inverse causal graph.
4. The target to optimize the causal graph.
5. The graph structure.
6. The natural gradient descent to optimize the model.
7. The NGD estimate the distribution of the model parameters during Python optimization.
8. Generative model to generate the inverse causal graph.
9. Double pendulum example.

Covariance and Causality

$$
\begin{aligned}
X_t &:= a_x X_{t-1} + a_{yx} Y_t + a_{zx} Z_t + \xi_t^x \\
Z_t &:= a_z Z_{t-1} + a_{xz} X_{t-1} + a_{yz} Y_t + a_{wz} W_{t-1} + \xi_t^z \\
Y_t &:= a_y Y_{t-1} + a_{xy} X_{t-1} + a_{wy} W_{t-2} + \xi_t^y \qquad (1) \\
W_t &:= a_w W_{t-1} + a_{yw} Y_t + a_{zw} Z_t + \xi_t^w \\
U_t &:= a_u U_{t-1} + a_{wu} W_t + \xi_t^u .
\end{aligned}
$$

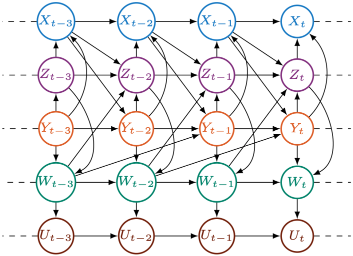


Figure 1: Running example. Left: Dynamic structural causal model (dynamic SCM). Right: Associated full-time causal graph $\mathcal{G}^{\mathrm{f}}$ (FTCG).

Summary Causal Graph

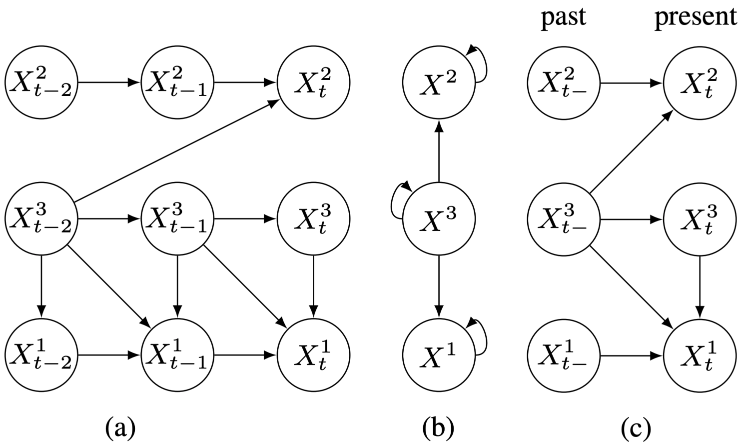


Figure 1: Example of a window causal graph (a), a summary causal graph (b) and an extended summary causal graph (c).

The dimensionality of the model

$$\begin{pmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,N} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,N} \\ \vdots & \vdots & w_{\kappa,\tau} & \vdots \\ w_{K,1} & w_{K,2} & \cdots & w_{K,N} \end{pmatrix} \underbrace{\begin{pmatrix} x_{t-1} & x_{t-2} & \cdots & x_{t-(N+T)} \\ x_{t-2} & x_{t-3} & \cdots & x_{t-(N+T-1)} \\ \vdots & \vdots & x_{t-\tau} & \vdots \\ x_{t-N} & x_{t-(N-1)} & \cdots & x_T \end{pmatrix}}_{\text{Hankel matrix } \mathrm{X}_{t-1}}.$$

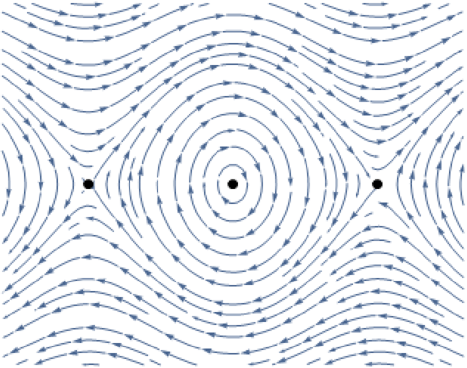
Convolution of f and g

$$(f * g)(t) = \int_{\tau=0}^{T} f(\tau)g(t-\tau)d\tau$$

The bilateral Laplace transform

$$F(s)\cdot G(s) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-st}\, f(u)\, g(t-u)\, \mathrm{d}u\, \mathrm{d}t = \int_{-\infty}^{\infty} e^{-st}(f*g)(t)\, \mathrm{d}t.$$

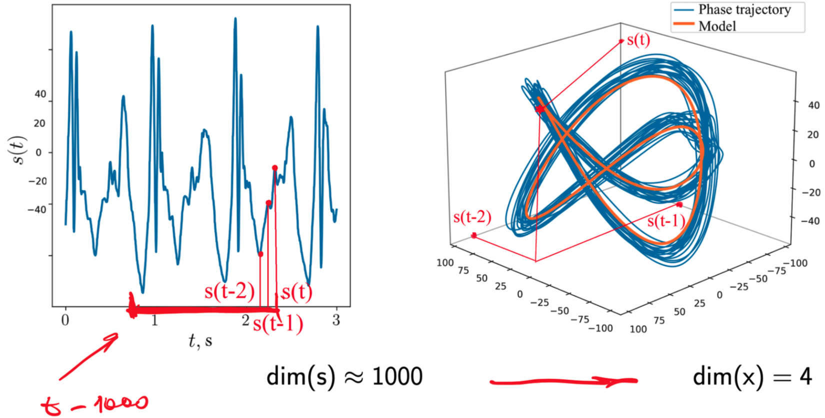
Phase Space of a pendulum: x,y-axis are $(\theta, \dot{\theta})$



$$\theta(t) = -\mu\theta(t) - \frac{g}{L}\sin\big(\theta(t)\big)$$

$$\frac{d}{dt}\begin{bmatrix}\theta(t)\\ \dot{\theta}(t)\end{bmatrix} = \begin{bmatrix}\dot{\theta}(t)\\ \ddot{\theta}(t)\end{bmatrix} = \begin{bmatrix}\dot{\theta}(t)\\ -\mu\dot{\theta}(t) - \frac{g}{L}\sin\big(\theta(t)\big)\end{bmatrix}$$

Phase trajectory



$\dim(s) \approx 1000 \longrightarrow \dim(x) = 4$

Reduce dimensionality with the principal component analysis, autoencoder $\boldsymbol{z} = \boldsymbol{W}^\mathsf{T}\boldsymbol{x}$ where $\boldsymbol{W}$ is an orthogonal (rotation) matrix. The first principal components are given by Singular Values Decomposition

$$\sqrt{\lambda_k}\boldsymbol{V_k} = \boldsymbol{X}^\mathsf{T}\boldsymbol{U}_k \quad \text{the SVD is} \quad \boldsymbol{X} = \boldsymbol{U}\Lambda\boldsymbol{V}^\mathsf{T}$$

$\mathrm{KL}\left(p(\mathrm{w}|X)\,\|\, p(\mathrm{w}|do(X=\mathrm{x})\right)$

$do(X_{t-p-2} = x_0)$

$(X_{t-p+1}, \underset{\substack{\shortparallel \\ x_0}}{X_{t-p-2}}, \ldots, X_{t-1}, X_t) \longrightarrow Y = X_{t+1}$

samples (training data)

$s^1 : (\quad) \quad (\quad)$

$s^2 : (\quad) \quad (\quad)$

$\vdots$

$s^n : (\quad) \quad (\quad)$

model: $Y = \sigma(WX)$

$W = \begin{bmatrix} w_1^T \\ \vdots \\ w_k^T \end{bmatrix}, \quad W \in \mathbb{R}^{k \times n_x}, \quad w_i \in \mathbb{R}^n_{1 \le i \le k}$

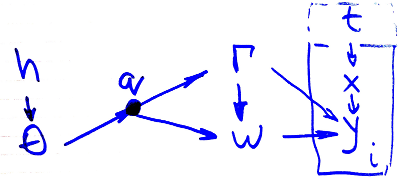
$X \in \mathbb{R}^{n_x \times 1}$

? $P(W|X)$

? $P(W|do(X = x_0))$

$KL\left(P(W|X)\,\|\,P(W|do(X = x_0))\right)$

Graphical model for the structure parameters Γ



Causal graph optimization

There given

1. Model structure Γ
2. Causal graph G
3. Causal hypergraph $\mathcal{G}$
4. Observable x and latent z spaces and data with reconstructions
5. Set of do-interventions in x and z

Set of criterions

$X \text{prind} Y$, $\mathrm{KL}(X||Y)$, $\mathrm{MI}(X, Y)$ and 3 rules of do-calculus

The algorithm

Forward (x $\rightarrow$ z) and backward (z $\rightarrow$ x)

1. Optimization step: get distributions (without do)
2. Select a (hyper)node
3. Run do-intervention
4. Optimization step: get distributions (with do)
5. Compute the criterion
6. Eliminate the $v \in \mathcal{G}$ and $\gamma \in \Gamma$
7. Continue until convergence